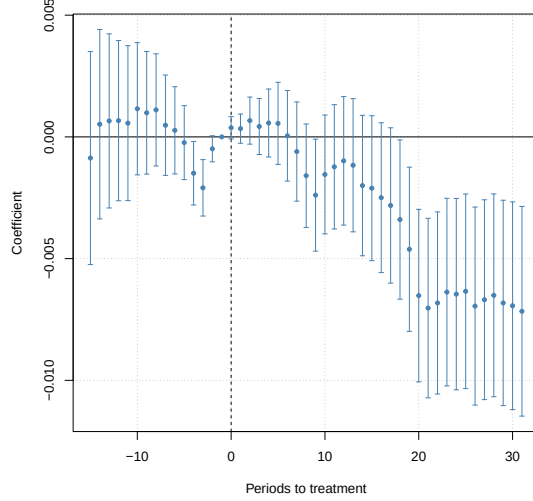


Figure 1: Effect of LLM exposure on firm-level jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the total number of employees in UK firms (log). Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. Vertical line indicates November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

show up in aggregate employment statistics. High and low LLM exposure exhibit similar employment trajectories prior to November 2022, which provides support for the validity of the parallel trends assumption. When averaged over the entire post-treatment period (Table 3), a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment, although the aggregate effect is only marginally significant due to the delayed onset of impacts.

We next examine how these aggregate effects vary across different types of workers. Figure 2 decomposes the effects between high- and low-seniority employees. Panel A shows junior positions decline immediately and statistically significantly, while Panel B demonstrates that senior positions initially remain stable before declining gradually, with the decrease only becoming significant after 20 months. Panel C shows that the timing difference in the emergence of effects leads to a temporary shift toward senior employees, although the compositional change attenuates over time, with the seniority distribution gradually reverting toward pre-treatment levels by the end of our observation period. All three panels show parallel pre-trends.

Table 3: Effect of LLM exposure on firm-level job outcomes

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	-0.003* (0.002)	-0.004** (0.002)	-0.002 (0.002)	-0.001** (0.0003)	106.313* (59.029)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, averaged over the post-treatment period. *Employees* is the total number of jobs in the firm. *Low seniority* and *High seniority* denote the number of positions in seniority levels 1–3 and 4–7, respectively. *Low seniority (%)* is the share of jobs that are low-seniority. *Av. comp.* is the mean annual salary plus bonuses. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level and reported in parentheses. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regression results in Table 3 show, averaged over the entire post-treatment period, a significant 0.4 percent reduction in junior employees per standard deviation increase in exposure, and a statistically insignificant reduction in senior employees. The fraction of junior employees goes down by 0.1 percentage points per standard deviation of LLM exposure ($p = 0.028$).

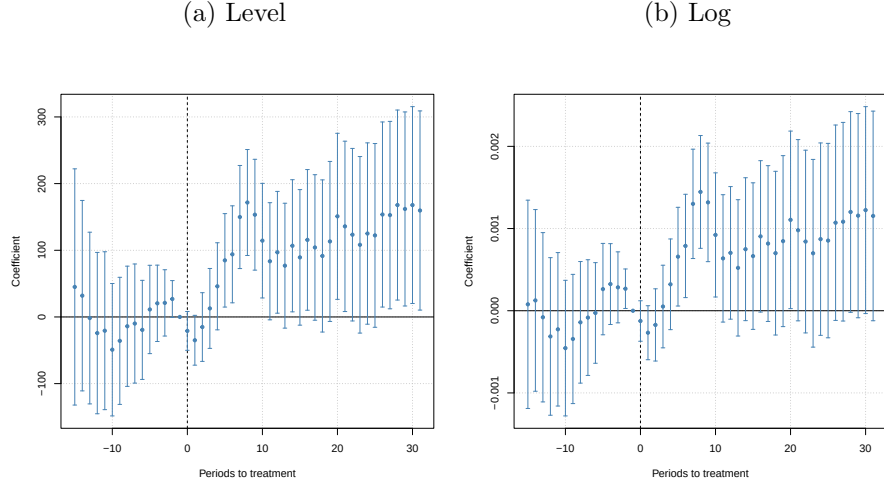
The observed compositional shift towards more senior roles has direct implications for compensation patterns. Figure 3 shows that average compensation rises modestly in highly exposed firms following the introduction of ChatGPT, with a one standard deviation increase in exposure associated with a £106 increase in average compensation, although the overall estimate is only marginally significant.

4.2 Job Posting Dynamics

4.2.1 Firm-Level Postings

While employment stocks adjust slowly, job postings provide a real-time window into firms’ evolving labor demand. We therefore examine how firms’ hiring intentions respond to LLM exposure by analyzing whether exposed firms post fewer vacancies overall and whether they adjust the composition of roles they seek to fill. This posting data captures the extensive margin of recruitment activity and reveals forward-looking decisions that precede the employment stock changes documented above. By tracking the same firms across both datasets, we can directly observe

Figure 3: Effect of LLM exposure on firm-level compensation



Notes: This figure plots event study estimates of the effect of LLM exposure on average compensation within UK firms. Compensation is defined as annual base salary plus bonuses. Panel A shows compensation in levels, Panel B in logs. Coefficients are estimated from Equation (3) and are shown relative to October 2022, the final pre-treatment month. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

Figure 4 presents event study estimates for the probability that exposed vs. less exposed firms post any job vacancy in a given month. The results show a change in hiring patterns following November 2022. Prior to this date, firms with different exposure levels did not have statistically significantly different trends in job postings.⁸ Following the release of ChatGPT, however, we observe that a one standard deviation increase in LLM exposure reduces firms' likelihood of posting new jobs by 1.0 percentage point ($p < 0.001$, see Table A1). This effect emerges gradually and becomes statistically significant after approximately 10 months, mirroring the timeline of employment stock adjustments.

Next, we decompose the aggregate hiring reduction by estimating occupation-specific effects across job categories. We estimate Equation (4) separately for 50 distinct occupational categories, with the outcome variable indicating whether a

⁸Although all pre-trend estimates are statistically insignificant, the point estimates exhibit a slight downward drift. If this reflects a differential pre-trend rather than sampling variation, the post-treatment estimates may overstate the true effect.

Table 4: Effect of LLM exposure on occupation-level job postings

	Job postings (log)	Salary	Salary (log)	Salary (SD)
LLM exposure x Post	-0.039** (0.018)	-495.483*** (135.270)	-0.011*** (0.003)	-56.802 (108.358)
Occupation fixed-effects	Yes	Yes	Yes	Yes
Year-by-month fixed effects	Yes	Yes	Yes	Yes
Observations	16,815	16,815	16,815	16,804
Adjusted R ²	0.962	0.948	0.953	0.729

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on occupation-level outcomes. *Job postings (log)* is the log number of new vacancies in occupation o in month t . *Salary* is the mean advertised annual compensation, *Salary (log)* is its logarithm, and *Salary (SD)* is the within-occupation standard deviation of posted salaries. *Exposure* is the occupation-level LLM exposure score from Eloundou et al. (2024). *Post* equals one for November 2022 onward. Standard errors are clustered at the occupation level. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2.2 Occupation-Level Postings

This analysis examines occupation-level job posting data to understand how aggregate labor demand evolves for specific roles. This analysis exploits variation in LLM exposure across 384 occupations, from minimally affected roles such as cooks (exposure score: 0) to highly exposed positions like proofreaders and copy markers (exposure score: 0.86). We examine three outcomes: posting volumes to measure aggregate demand, advertised salaries to assess compensation responses, and within-occupation salary dispersion to measure wage polarization.

Figure 6 presents event study estimates for job posting volumes. Prior to November 2022, high- and low-exposure occupations follow parallel trajectories, validating our identification strategy. Following ChatGPT’s release, however, exposed occupations experience a sustained reduction in hiring demand, with effects first becoming statistically significant after 7 months and intensifying thereafter. After 20 months, we see a slight resurgence, and job postings converge back to their original levels. When averaged across the post-treatment period (Table 4), a one standard deviation increase in LLM exposure is associated with a 3.9% reduction in posting volume ($p = 0.029$). This is an economically meaningful change.

Turning to compensation, we observe immediate and persistent wage pressure for exposed occupations. Figure 7 indicates that posted salaries begin declining shortly after ChatGPT’s introduction. A one standard deviation increase in LLM exposure is associated with a £495 decline in posted salaries and a 1.1% reduction in log